Whitepaper: Modern Cloud Data Architecture

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Data platforms have evolved from monolithic repositories into decentralized, lakehouse-driven ecosystems. Modern architectures emphasize modularity, zero-copy querying, and governance at scale.

A primary tenet of the modern lakehouse is the separation of compute and storage. Open table formats such as Apache Iceberg provide ACID transactions, schema evolution, and partition pruning across diverse execution engines.

Furthermore, security and governance are no longer bolted on as afterthoughts. Fine-grained row- and column-level access controls, column masking, and automated lineage tracking ensure compliance with strict international regulations such as GDPR and CCPA.